

Daphne Wilkerson

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Summary

I am a self-motivated music software designer and engineer excited to join a team similarly interested in creating cutting-edge digital tools.

Education

Stanford University Stanford, CA
BS in Computer Science (Concentration in Human Computer Interaction), Minor in Music June 2023
GPA: 3.99

Coursework focus: team-based software development, DSP, design principles & critique, prototyping, technical writing, and usability testing

Coronado High School Coronado, CA
Valedictorian — GPA: 4.46 June 2018
Coronado School of the Arts, Piano Performance

Experience

Armory—Independent Music Software Design June 2023 – Present

Released *TurnTableSaw*, a plugin instrument for turntable scratching composition and live performance, designed with Blender and Photoshop and built with JUCE/C++ over a nine month period

Released *LaunchCodes*, a plugin for synthesized explosive sound, built with JUCE over a five month period

Designed several unique and themed user interfaces with dozens of original GUI components such as a zoomable B-spline double-envelope editor with multi-node selection & several editing modes, multi-pole filters, and interactive waveform views

Engineered solutions to several DSP issues including anti-aliasing samples at arbitrary rates and simultaneous discontinuity smoothing & DC offset challenges, as well as developed several DSP effects such as distortion units and delay lines

Developed for Mac and Windows using Xcode and Visual Studio Code—debugged, profiled, and refined the performance GUI and DSP processes

Managed branding, media, and software distribution of my products, and created [a highly interactive website](#) from scratch to feature my work and share my design processes

Stanford University—Research in Interaction Design January 2022 – June 2023

Conducted research on touch gestures for audio-based interaction over several months with three co-researchers, involving ten participants and producing a paper on our findings (unpublished, available [here](#))

Oversaw design work of four high-fidelity mobile and web app prototypes in teams of four developers using React and Figma, in domains such as music education, music-sharing, and artificial intelligence

Won best overall app prototype in a design contest of over 50 group competitors

Stanford Taiko—Artistic Director September 2021 – June 2022

Oversaw twice-weekly rehearsals for nationally respected Japanese drumming ensemble of 15 members

Updated and maintained organization website with university domain access

Skills

Programming: C++, C, JavaScript, HTML/CSS, Python, Git

Design: Figma, Illustrator, Blender, Photoshop, React | Wireframing, Prototyping, User Testing

Music Experience

Music production (Ableton, Logic), studio recording

DSP algorithm design (C++, Chuck, Faust, Max, Web Audio) and music visualization (C++, Processing)

Language Proficiency in French, German, and Japanese

Hobbies include fashion history, horror films, music composition, and gaming